Number theory – 2

Orders and primitive roots

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Theoretical minimum. The multiplicative group of residues is the engine of elementary number theory: almost every divisibility statement about $a^n \pm 1$ is a statement about the order of a.

Definitions. For $\gcd(a, n) = 1$ the *order* $\operatorname{ord}_n(a)$ is the least $d > 0$ with $a^d \equiv 1 \pmod{n}$. A *primitive root* modulo n is an element of order $\varphi(n)$, that is a generator of $(\mathbb{Z}/n\mathbb{Z})^*$.

Theorem 1 (the basic property). $a^m \equiv 1 \pmod{n}$ if and only if $\operatorname{ord}_n(a) \mid m$; in particular the order divides $\varphi(n)$. *Proof:* divide m by the order with remainder. *Corollary:* if $p \mid a^n - 1$ then $\operatorname{ord}_p(a) \mid n$ and $\operatorname{ord}_p(a) \mid p - 1$, so every prime divisor of $a^n - 1$ either divides some $a^d - 1$ with $d \mid n$, $d < n$, or is congruent to 1 modulo n. This one corollary settles most olympiad questions about the prime divisors of $2^n - 1$ and of Fermat numbers.

Theorem 2 (primitive roots modulo a prime). For every prime p the group $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic. *Proof:* for $d \mid p - 1$ the congruence $x^d = 1$ has at most d roots, since a polynomial of degree d over the field $\mathbb{F}_p$ has at most d roots; hence the number of elements of order exactly d is either 0 or $\varphi(d)$; summing $\sum_{d \mid p-1} \varphi(d) = p - 1$ forces it to be $\varphi(d)$ for every d, and in particular $\varphi(p - 1) > 0$ elements of order $p - 1$ exist.

Theorem 3 (the general structure). $(\mathbb{Z}/n\mathbb{Z})^*$ is cyclic exactly for $n = 1, 2, 4, p^k, 2p^k$ with p an odd prime; for $n = 2^k$, $k \geqslant 3$, it is $\mathbb{Z}/2 \times \mathbb{Z}/2^{k-2}$. In general the Chinese remainder theorem gives $(\mathbb{Z}/n\mathbb{Z})^* \cong \prod (\mathbb{Z}/p_i^{k_i}\mathbb{Z})^*$, so the exponent of the group is lcm of the individual orders — the Carmichael function $\lambda(n)$, which can be much smaller than $\varphi(n)$.

Fact 1 (index calculus). Fixing a primitive root g modulo p turns multiplication into addition: every residue is g^k with k well defined modulo

$p-1$, and $x^m \equiv a$ becomes a linear congruence in the index. In particular a is an m-th power modulo p if and only if $a^{(p-1)/\gcd(m,p-1)} \equiv 1$, and the m-th powers form a subgroup of index $\gcd(m, p-1)$.

Fact 2 (Zsygmondy). For $a > b \geqslant 1$ coprime and $n \geqslant 2$ the number $a^n - b^n$ has a prime divisor dividing no $a^k - b^k$ with $k < n$, apart from the exceptions $(a, b, n) = (2, 1, 6)$ and $a + b$ a power of two with $n = 2$. The statement may be quoted; it converts many existence questions into a one-line check.

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1. Prove Theorem 1 and its corollary, and deduce: every prime divisor of $2^p - 1$ with p prime is congruent to 1 modulo $2p$; for $k \geqslant 2$ every prime divisor of the Fermat number $2^{2^k} + 1$ is congruent to 1 modulo 2^{k+2}; and there are infinitely many primes congruent to 1 modulo n for every fixed n.

2. Prove that $(\mathbb{Z}/p\mathbb{Z})^*$ is cyclic, counting the elements of each order, and determine for which n a primitive root modulo n exists. Then show that 2 is a primitive root modulo 3^k for every k, and find the order of 2 modulo $2^k - 1$.

3. Let $p > 3$ be a prime. A sequence $a_1, \ldots, a_{p-1}$ of integers is *wonky* if the terms are distinct modulo p, the partial sums a_1a_2, $a_1a_2 + a_2a_3$, $\ldots$, $a_1a_2 + a_2a_3 + \cdots + a_{p-1}a_1$ are distinct modulo p, and $a_ia_{i+2} \not\equiv a_{i+1}^2 \pmod{p}$ for all i, indices taken cyclically. Determine for which p a wonky sequence exists. (ICMC)

4. Let n be an integer and p a prime greater than 2. Show that

$$(p-1)^n\, n! \;\Big|\; (p^n - 1)(p^n - p)(p^n - p^2) \cdots (p^n - p^{n-1}),$$

   and explain the right-hand side as the order of $\mathrm{GL}_n(\mathbb{F}_p)$ (Abstract algebra – 4). (CIIM 2014)

5. Find all primes $p > 5$ for which there exist an integer a and an integer r with $1 \leqslant r \leqslant p - 1$ such that the sequence $1, a, a^2, \ldots, a^{p-5}$ can be rearranged into a sequence $b_0, b_1, \ldots, b_{p-5}$ with $b_n - b_{n-1} - r$ divisible by p for $1 \leqslant n \leqslant p - 5$. (Putnam 2024)